

Jian Xu

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✧ EDUCATION and WORK EXPERIENCE

Postdoc researcher	McGill University	2025/09 – present
Ph.D.	University of Science and Technology of China	2021/09 – 2025/07
Visiting Ph.D. Student	McGill University	2023/11 – 2024/11
M.S.	University of Science and Technology of China	2019/09 – 2021/08
B.S.	University of Science and Technology of China	2015/09 – 2019/08

✧ RESEARCH INTEREST

Improve assessment of induced seismic hazards, by:

- Tracking high-pressure fluids via time-lapse in-situ Vp/Vs techniques
- Constructing tectonic structures via high-resolution velocity model

Deepen understanding of earthquake source physics, by:

- Estimating high-resolution earthquake source properties with dense seismic arrays
- Comparing and integrating source properties of induced and/or natural earthquakes

✧ PUBLICATION (# co-first Author)

- (1) **Xu, J.**, Liu, Y., Li J. In-Situ Vp/Vs reveals fluid accumulation triggers the large induced earthquakes in the Changning shale gas field, southern Sichuan basin, **in preparation**.
- (2) Li, J., **Xu, J.**[#], Tan, Y. Yang, W., Lu, R., Jiang, G., Ma, X., Hu, J., Zhang, F., Wang, R., Tang, J. & Zhang, H. Review of a decade of induced earthquakes in the Changning shale gas field, Sichuan Basin of China, **in preparation**.
- (3) **Xu, J.**, Li, J., Yang, W., Chen, G., Liu, Y., Verdecchia, A., Harrington, R. M., Lu, R., Tan, Y., Ye Y. & Tang, J. (2025). Shallow Lingering and Deep Transient Seismicity Related to Hydraulic Fracturing in the Changning Shale Gas Field, Sichuan Basin, China. *Journal of Geophysical Research: Solid Earth*, 130(4), e2024JB030279.

- (4) Li, J., **Xu, J.**, Zhang, H., Yang, W., Tan, Y., Zhang, F., Meng, L., Zang, Y., Miso, S., Guo, C., Li, Z., Lu, R. & Sun, J. (2023). High seismic velocity structures control moderate to strong induced earthquake behaviors by shale gas development. *Communications Earth & Environment*, 4(1), 188.
- (5) Zhao, Y., Li, J., **Xu, J.**, Yao, H., Zhu, G., Yang, H., Zhang, J. & Lu, R. (2023). High-resolution velocity structure and seismogenic potential of strong earthquakes in the Bamei-Kangding segment of the Xianshuihe fault zone. *Science China Earth Sciences*, 66(9), 1960-1978.
- (6) Tan, Y., Qian, J., Hu, J., Zhang, H., Xing, H., Li, J., **Xu, J.**, Yang, W., Gu, N. & Miao, S. (2023). Tomographic evidences for hydraulic fracturing induced seismicity in the Changning shale gas field, southern Sichuan Basin, China. *Earth and Planetary Science Letters*, 605, 118021.
- (7) **Xu, J.**, Wang, K., Li, J., Gu, N., Ding, N., Qian, J., Yang, W. & Zhang, H. (2021). High resolution tomography of the Tanlu fault zone near Hefei with passive seismic and magnetotelluric linear array data. *Earthquake Science*, 34(1), 24-35.

✧ PRESENTATION

Oral:

Xu, J., Liu, Y., Li, J. & Yang W. Spatiotemporal variations of in-situ Vp/Vs ratios reveal the moderate to strong induced earthquakes are triggered by fluid accumulation in the Changning shale gas field. *Chinese Geoscience Union*, Xiamen City, 2024 (in Chinese).

Xu, J., Yang, W., Chen, G., Zhang, Z., & Li, J. Two Distinct Patterns of Hydraulic Fracturing-Induced Seismicity in the Changning Shale Gas Field, Southern Sichuan basin, China. *AGU Fall Meeting, San Francisco, 2023*.

Xu, J., Li, J., Zhang, H., Yang, W., Tan, Y., Guo, C. & Miao, S. High-velocity structure controls the earthquake behaviors of moderate to strong events in the Changning shale gas field. *Chinese Geoscience Union*, Zhuhai City, 2023 (**Outstanding Student Presentation Award, in Chinese**).

Xu, J., Yang, W. & Li, J. High-precision earthquake relocation reveals reactivations of a 3-D fracture network: insight from a cluster of hydraulic fracturing induced seismicity in Changning shale gas field. *Chinese Geoscience Union*, Fuzhou City, 2022. (On-line, in Chinese)

Poster:

Xu, J., Li, J., Zhang, H., Yang, W., Tan, Y., Guo, C., ... & Li, Z. Earthquake behaviors for moderate to strong induced events are controlled by high-velocity bodies near shale reservoirs in southern Sichuan basin, China. *EGU General Assembly Conference, Vienna, 2024*.

Xu, J., Li, J., Zhang, H., Yang, W. Miao, S. & Tan, Y. High seismic velocity structures control moderate to strong induced earthquake behaviors by shale gas development. *2023 International Professional Association of Chinese Earth Scientists annual meeting*. (**Second Prize in Poster Presentation**)

Xu, J., Li, J., Zhang, H. & Yang, W. High-low seismic velocity transition zones control hydraulic fracturing-induced earthquake behaviors. *6th China Workshop on Dense Seismic Arrays*, Shenzhen, 2023.

✧ PEER-REVIEW ACTIVITY

Journal of Geophysical Research: Solid Earth

Reviewer and Co-reviewer

Tectonophysics

Reviewer

Acta Seismologica Sinica (Chinese Version)

Reviewer

✧ AWARD

First-Class Academic Excellence Scholarship, USTC	2019-2025
Outstanding Student Presentation Award, Chinese Geoscience Union	2023
Chinese Government Scholarship, China Scholarship Council	2023
Outstanding Graduate Award, USTC	2019
National Encouragement Scholarship, Ministry of Education, China	2017

✧ EXPERIENCE

Seismic Network Design:

Extensive experience in designing seismic networks in a variety of geological settings:

~300 stations in the Sichuan Basin for monitoring hydraulic fracturing-induced seismicity

~400 stations in the high-elevation Sichuan–Yunnan seismic zone for monitoring tectonic earthquakes

~400 stations along the Tanlu Fault Zone in East China for crustal structure tomography

Seismic Station Installation and Maintenance:

May & November 2024 – Maintenance of broadband seismic stations near the St. Lawrence River, Eastern Canada

June 2023 – Deployment and maintenance of 4G real-time seismic stations in the Sichuan Basin, China

August 2021 – Retrieval of short-period seismic stations in the Tanlu Fault Zone, China

March 2021 – Seismic data acquisition in the Sichuan Basin, China

Student Advisor and Class Mentor (non-academic):

Sept 2019 – Aug 2023, Advisor for a class of 68 undergraduate students, providing continuous support throughout their four-year program in areas including academic progress, mental health, scholarship applications, graduate school preparation, and job placement.

President of the Student Association (non-academic)

May 2017 – April 2018, School of Earth and Space Sciences, USTC